Understanding Docker Volumes

Suppose there's an application, where you want to store the data permanently in some file, then how to handle these datas.

Created a python code:

python program which asks to enter usernames in one line seperated by space. after user enters the name, it asks to should I show all the names you have provided now and in previous execution. meaning it stores all the names and the time at which it is added into a text file. shows y/n options. then asks should I delete the memory file that stores name and the time name is added. shows y/n options. if this is one cycle, make it as the infinite loop until the user wants quit or press q to quit, else continues the loop.

Code:

```
import os
from datetime import datetime

FILE_NAME = "user_memory.txt"

def add_usernames():
    usernames = input("Enter usernames separated by space (or press q to quit): ").strip()

    if usernames.lower() == 'q':
        return False

    names_list = usernames.split()
    current_time = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

    with open(FILE_NAME, "a") as file:
        for name in names_list:
            file.write(f"{name} | {current_time}\n")

    return True


def show_all_names():
    if not os.path.exists(FILE_NAME):
        print("\nNo data found yet.\n")
        return

    print("\nStored Usernames with Timestamps:")
    print("----------------------------------")

    with open(FILE_NAME, "r") as file:
        data = file.read()
```

```
        print(data if data else "No records found.")

    print("----------------------------------\n")


def delete_memory():
    if os.path.exists(FILE_NAME):
        os.remove(FILE_NAME)
        print("\nMemory file deleted successfully.\n")
    else:
        print("\nNo memory file exists.\n")


# 🔁 Infinite Loop
while True:
    if not add_usernames():
        print("\nExiting program... 👋")
        break

    # Show stored names
    choice = input("Show all names (current + previous)? (y/n): ").strip().lower()
    if choice == 'y':
        show_all_names()

    # Delete memory file
    choice = input("Delete memory file? (y/n): ").strip().lower()
    if choice == 'y':
        delete_memory()

    print("\n--- Next Cycle ---\n")
```

Output in Terminal:

```
C:\Users\kiran\OneDrive\Desktop\Python Code>py PyCode.py
Enter usernames separated by space (or press q to quit): routh kiran babu
Show all names (current + previous)? (y/n): y

Stored Usernames with Timestamps:
----------------------------------
routh | 2026-03-22 15:11:42
kiran | 2026-03-22 15:11:42
babu | 2026-03-22 15:11:42

----------------------------------

Delete memory file? (y/n): n
```

```
--- Next Cycle ---

Enter usernames separated by space (or press q to quit): routh1 kiran1
Show all names (current + previous)? (y/n): y

Stored Usernames with Timestamps:
----------------------------------
routh | 2026-03-22 15:11:42
kiran | 2026-03-22 15:11:42
babu | 2026-03-22 15:11:42
routh1 | 2026-03-22 15:12:00
kiran1 | 2026-03-22 15:12:00

----------------------------------

Delete memory file? (y/n): n

--- Next Cycle ---

Enter usernames separated by space (or press q to quit): babu1
Show all names (current + previous)? (y/n): y

Stored Usernames with Timestamps:
----------------------------------
routh | 2026-03-22 15:11:42
kiran | 2026-03-22 15:11:42
babu | 2026-03-22 15:11:42
routh1 | 2026-03-22 15:12:00
kiran1 | 2026-03-22 15:12:00
babu1 | 2026-03-22 15:12:14

----------------------------------

Delete memory file? (y/n): n

--- Next Cycle ---

Enter usernames separated by space (or press q to quit): q

Exiting program... 👋

C:\Users\kiran\OneDrive\Desktop\Python Code>
```

Folder structure:

Name	Date modified
PyCode.py	22-Mar-26 3:02 PM
user_memory.txt	22-Mar-26 3:12 PM

Memory file has:

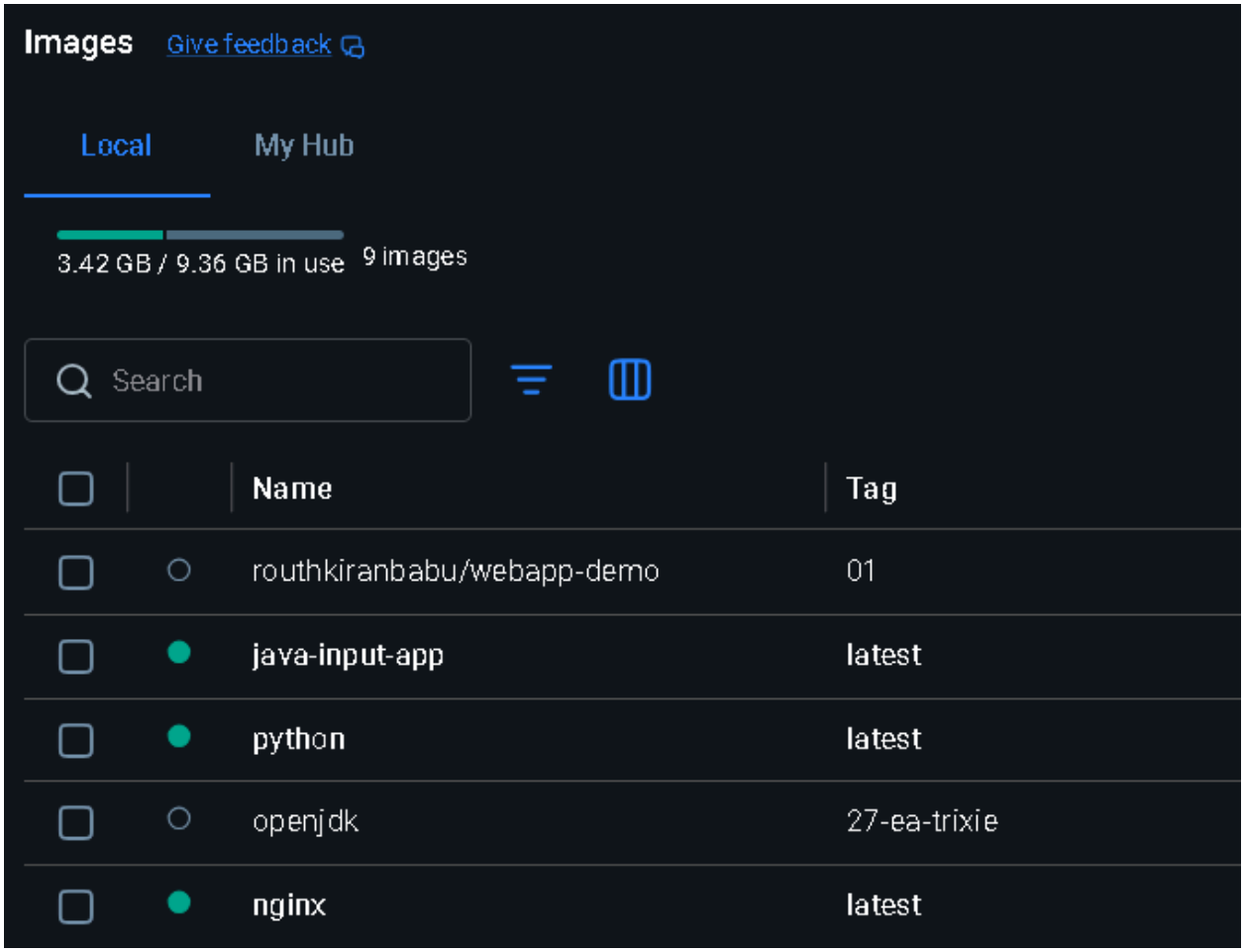
```
routh | 2026-03-22 15:11:42
kiran | 2026-03-22 15:11:42
babu | 2026-03-22 15:11:42
routh1 | 2026-03-22 15:12:00
kiran1 | 2026-03-22 15:12:00
babu1 | 2026-03-22 15:12:14
```

Now how to handle this in docker.
Created a Docker file:

```
FROM python

WORKDIR /myapp

COPY ./PyCode.py .

CMD [ "python", "PyCode.py" ]
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker build .
ERROR: failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine;
ngine: The system cannot find the file specified.
```

To remove this error: Open the Docker Desktop.



Then try again

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker build .
ERROR: failed to connect to the docker API at npipe:////./pipe/
ngine: The system cannot find the file specified.
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker build .
[+] Building 3.4s (8/8) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 121B
 => [internal] load metadata for docker.io/library/python:lates
 => [internal] load .dockerignore
 => => transferring context: 2B
```

In docker desktop:

Name	Tag	Image ID	Created
routhkiranbabu/webapp-demo	01	7023f47e1922	4 days ago
java-input-app	latest	2ee3d21b2faa	4 days ago
python	latest	0040600478fe	5 days ago
openjdk	27-ea-trixie	4a7da6fcd5de	5 days ago
nginx	latest	dec7a90bd097	5 days ago
mywebapp	02	b90c1a14ca28	6 days ago
routhkiranbabu/webapp-demo	02	b90c1a14ca28	6 days ago
mywebapp	01	89eacf391531	6 days ago
<none>	<none>	2ca300af969a	6 days ago
<none>	<none>	d209b8ea96db	1 minute ago

In terminal

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker images -a

IMAGE                          ID            DISK USAGE   CONTENT SIZE   EXTRA
java-input-app:latest          2ee3d21b2faa       636MB         198MB    U
mywebapp:01                    89eacf391531      2.11GB         491MB    U
mywebapp:02                    b90c1a14ca28      2.11GB         491MB
nginx:latest                   dec7a90bd097       240MB        65.8MB    U
openjdk:27-ea-trixie           4a7da6fcd5de      1.25GB         396MB
python:latest                  0040600478fe      1.63GB         432MB    U
routhkiranbabu/webapp-demo:01  7023f47e1922      2.11GB         491MB
routhkiranbabu/webapp-demo:02  b90c1a14ca28      2.11GB         491MB
<untagged>                     d209b8ea96db      1.61GB         414MB
<untagged>                     2ca300af969a      2.11GB         491MB    U
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

Lets run the image, since it is interactive use **-it, –rm** is used when the container store, it must automatically remove. No need to use more command to remove. Provide a name using –name:

```
\Python Code> docker run -it --rm --name mypythonapp d209b8ea96db
```

Output:

```
Enter usernames separated by space (or press q to quit): routh kiran babu
Show all names (current + previous)? (y/n): y

Stored Usernames with Timestamps:
----------------------------------
routh | 2026-03-22 09:59:36
kiran | 2026-03-22 09:59:36
babu | 2026-03-22 09:59:36

----------------------------------

Delete memory file? (y/n): n

--- Next Cycle ---

Enter usernames separated by space (or press q to quit): babu
Show all names (current + previous)? (y/n): y

Stored Usernames with Timestamps:
----------------------------------
routh | 2026-03-22 09:59:36
kiran | 2026-03-22 09:59:36
babu | 2026-03-22 09:59:36
babu | 2026-03-22 09:59:51

----------------------------------

Delete memory file? (y/n): n

--- Next Cycle ---
```

```
Enter usernames separated by space (or press q to quit): name1 name2
Show all names (current + previous)? (y/n): y

Stored Usernames with Timestamps:
----------------------------------
routh | 2026-03-22 09:59:36
kiran | 2026-03-22 09:59:36
babu | 2026-03-22 09:59:36
babu | 2026-03-22 09:59:51
name1 | 2026-03-22 10:00:12
name2 | 2026-03-22 10:00:12

----------------------------------

Delete memory file? (y/n): n

--- Next Cycle ---

Enter usernames separated by space (or press q to quit): q

Exiting program... 👋
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

Inside the container it creates a file. Python program running in the isolated environment. Since program is stop and so the container gets removed due to **–rm.** And so file is removed.
To check the containers:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

Now let's add volume:
-v for volume, name of the volume as **myvolume:path_to_store** path_to_store meaning to provide a path where you want to store the data, since **WORKDIR /myapp** so path_to_store is /myapp/

```
Dockerfile
Dockerfile > ...
1  FROM python

3  WORKDIR /myapp
```

so

```
\Python Code> docker run -it --rm -v myvolume:/myapp/ d209b8ea96db
 (or press q to quit):
```

Output:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker run -it --rm -v myv
Enter usernames separated by space (or press q to quit): routh kiran babu
Show all names (current + previous)? (y/n): y

Stored Usernames with Timestamps:
----------------------------------
routh | 2026-03-22 12:22:19
kiran | 2026-03-22 12:22:19
babu | 2026-03-22 12:22:19

----------------------------------

Delete memory file? (y/n): n

--- Next Cycle ---

Enter usernames separated by space (or press q to quit): name1 name2
Show all names (current + previous)? (y/n): y

Stored Usernames with Timestamps:
----------------------------------
routh | 2026-03-22 12:22:19
kiran | 2026-03-22 12:22:19
babu | 2026-03-22 12:22:19
name1 | 2026-03-22 12:22:36
name2 | 2026-03-22 12:22:36

----------------------------------

Delete memory file? (y/n): n

--- Next Cycle ---

Enter usernames separated by space (or press q to quit): q

Exiting program... 👋
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker run -it --rm -v myv
```

IN the next run:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker run -it --rm -v myvolume:/myapp/ d209b8ea96db
```

Result:

```
Enter usernames separated by space (or press q to quit): name3
Show all names (current + previous)? (y/n): y

Stored Usernames with Timestamps:
----------------------------------
routh | 2026-03-22 12:22:19
kiran | 2026-03-22 12:22:19
babu | 2026-03-22 12:22:19
name1 | 2026-03-22 12:22:36
name2 | 2026-03-22 12:22:36
name3 | 2026-03-22 12:23:23

----------------------------------

Delete memory file? (y/n):
```

Meaning it save the data, even if container is deleted using –rm.

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

To know more above docker volume:

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker volume --help
Usage:  docker volume COMMAND

Manage volumes

Commands:
  create      Create a volume
  inspect     Display detailed information on one or more volumes
  ls          List volumes
  prune       Remove unused local volumes
  rm          Remove one or more volumes

Run 'docker volume COMMAND --help' for more information on a command.
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker volume ls
DRIVER    VOLUME NAME
local     myvolume
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```

Means this volume is locally present.

```
PS C:\Users\kiran\OneDrive\Desktop\Python Code> docker volume inspect myvolume
[
    {
        "CreatedAt": "2026-03-22T12:21:26Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/myvolume/_data",
        "Name": "myvolume",
        "Options": null,
        "Scope": "local"
    }
]
PS C:\Users\kiran\OneDrive\Desktop\Python Code>
```